APM1205 — Summative Assessment 1

Comprehensive Linear Modeling Analysis

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Student name: _____

Course and section: _____

Executive Summary

This report investigates hourly Seoul bike-sharing demand using simple and multiple linear regression, statistical inference, matrix least squares, categorical predictors, ANOVA, interaction analysis, and model diagnostics. The findings describe associations in the observed dataset and should not be interpreted as causal effects.

Research question: How are weather conditions, time of day, season, holidays, and bike-service availability associated with hourly bike rental demand in Seoul?

A. Research Question and Data Preparation

Real-world problem

Bike-sharing operators need to understand how rental demand varies with weather, time, season, holidays, and service availability. This analysis estimates those relationships to support descriptive planning and demand-management decisions.

Research hypotheses

• H1: Mean rental demand differs with temperature.

• H2: At least one selected quantitative predictor is associated with rental demand after adjusting for other predictors.

• H3: Mean rental demand differs across seasons.

• H4: The association between season and rental demand depends on whether the bike service is functioning.

Dataset and measurement scales

The data file contains 8760 rows before cleaning and 14 columns. The analysis uses hourly rental count as the quantitative response. Temperature, humidity, wind speed, visibility, dew-point temperature, solar radiation, rainfall, snowfall, and hour are quantitative predictors when available. Season, holiday, functioning day, and weather are treated as categorical predictors when present. Rental count is a quantitative count variable; hour is numeric but represents a cyclical time-of-day measure; categorical labels are nominal factors.

Cleaning removed 0 duplicate rows and excluded rows with missing rental counts. Missing values in predictors were handled by complete-case analysis for the relevant model. The complete-case multiple model used 8760 observations. See the data-quality and cleaning CSV tables for details.

Table 1. Data cleaning summary

Measure	Value
Original rows	8760
Rows after duplicate removal and response check	8760
Duplicate rows detected	0
Missing cells before cleaning	0
Rows used in the main complete-case model	8760

Table 2. Variable types and missing values

Variable	Data_Type	Missing_Count	Missing_Percent	Unique_Values
date	datetime64[ns]	0	0	365
rented_bike_count	int64	0	0	2166
hour	int64	0	0	24

Variable	Data_Type	Missing_Count	Missing_Percent	Unique_Values
temperature	float64	0	0	546
humidity	int64	0	0	90
wind_speed	float64	0	0	65
visibility	int64	0	0	1789
dew_point_temperature	float64	0	0	556
solar_radiation	float64	0	0	345
rainfall	float64	0	0	61
snowfall	float64	0	0	51
seasons	category	0	0	4
holiday	category	0	0	2
functioning_day	category	0	0	2

B. Exploratory Data Analysis

Descriptive statistics summarize the response and quantitative predictors. Graphs are used to inspect skewness, potential outliers, relationships, and differences between groups. A linear model may be sensitive to skewed counts, nonlinear relationships, and unequal variance; these issues are revisited in the diagnostics section.

Table 3. Descriptive statistics

Variable	count	mean	std	min	25%	50%	75%	max	missing_in_cle
rented_bike_coun	8760	704.6	645	0	191	504.5	1065	3556	0
temperature	8760	12.88	11.94	-17.8	3.5	13.7	22.5	39.4	0
humidity	8760	58.23	20.36	0	42	57	74	98	0
wind_speed	8760	1.725	1.036	0	0.9	1.5	2.3	7.4	0
solar_radiation	8760	0.5691	0.8687	0	0	0.01	0.93	3.52	0
rainfall	8760	0.1487	1.128	0	0	0	0	35	0
snowfall	8760	0.07507	0.4367	0	0	0	0	8.8	0
hour	8760	11.5	6.923	0	5.75	11.5	17.25	23	0
visibility	8760	1437	608.3	27	940	1698	2000	2000	0
dew_point_tempe	8760	4.074	13.06	-30.6	-4.7	5.1	14.8	27.2	0

Figure 1. Distribution of hourly rental counts.

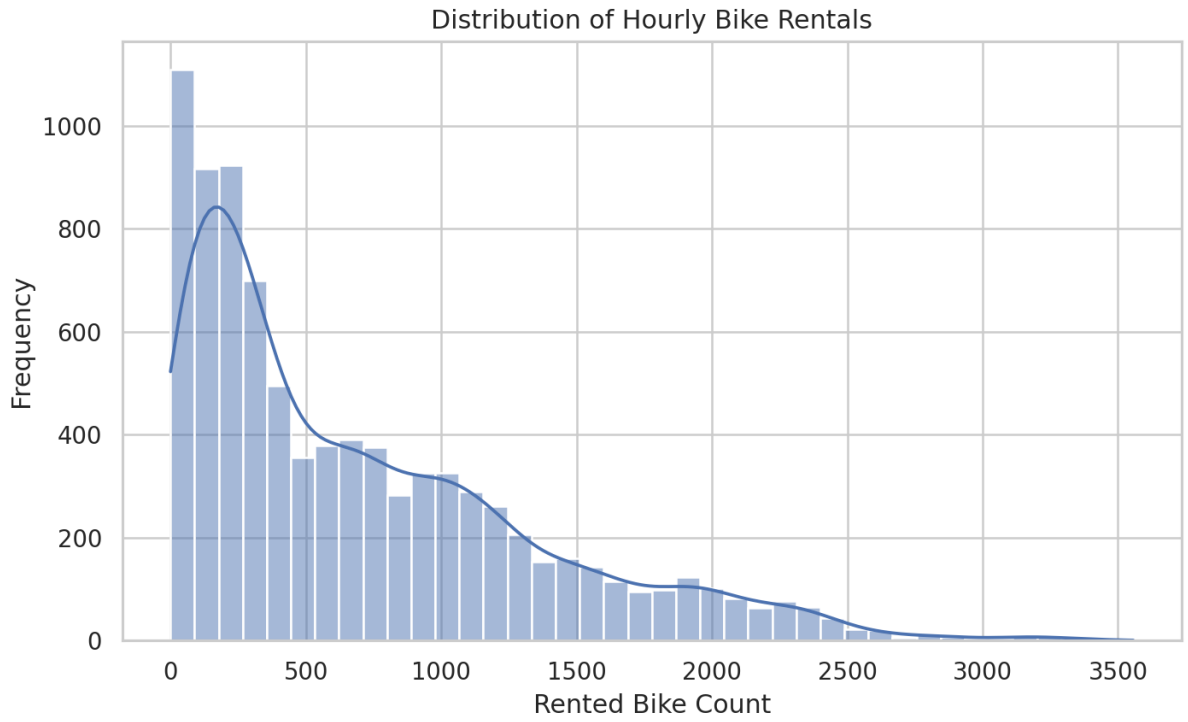


Figure 2. Rental counts versus temperature with a fitted straight line.

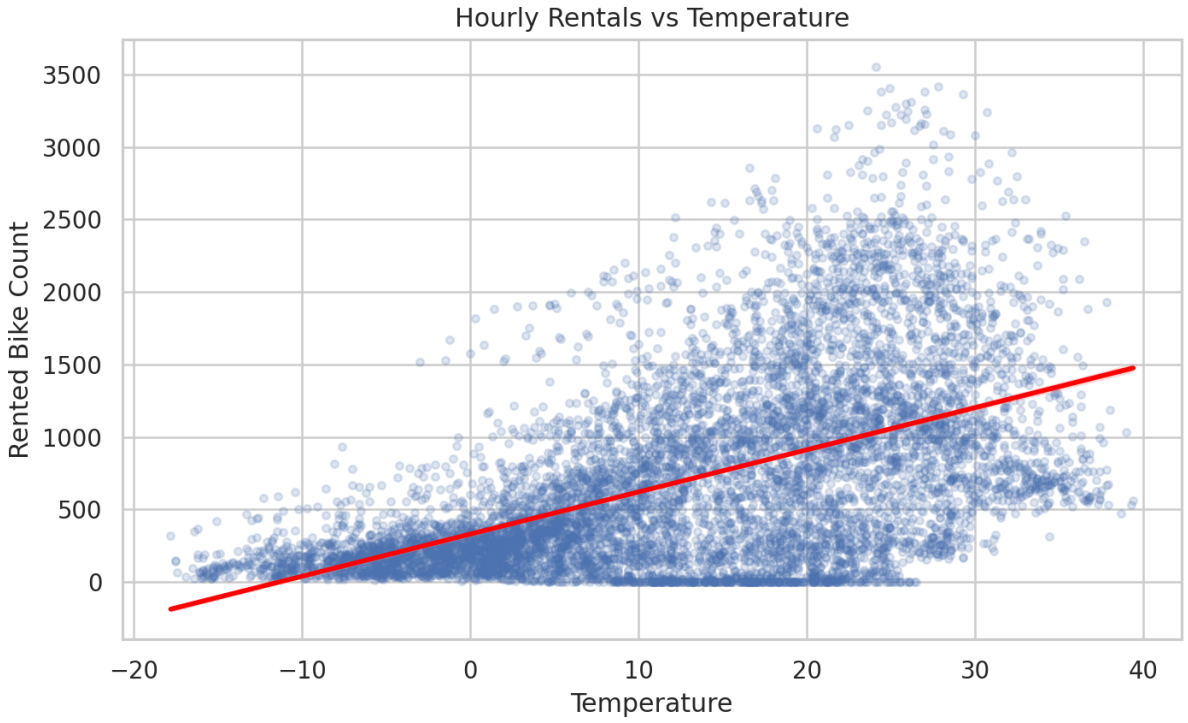


Figure 3. Correlations among quantitative variables.

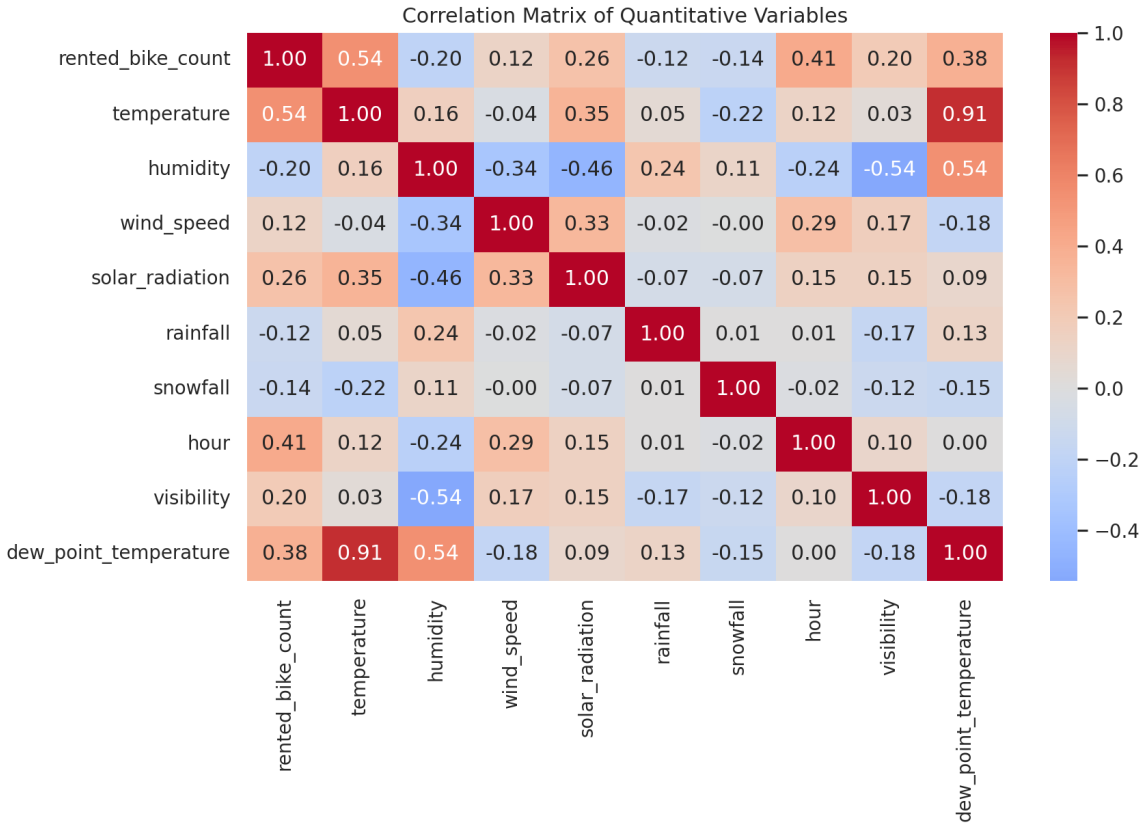


Figure 4. Rental-count distributions across seasons.

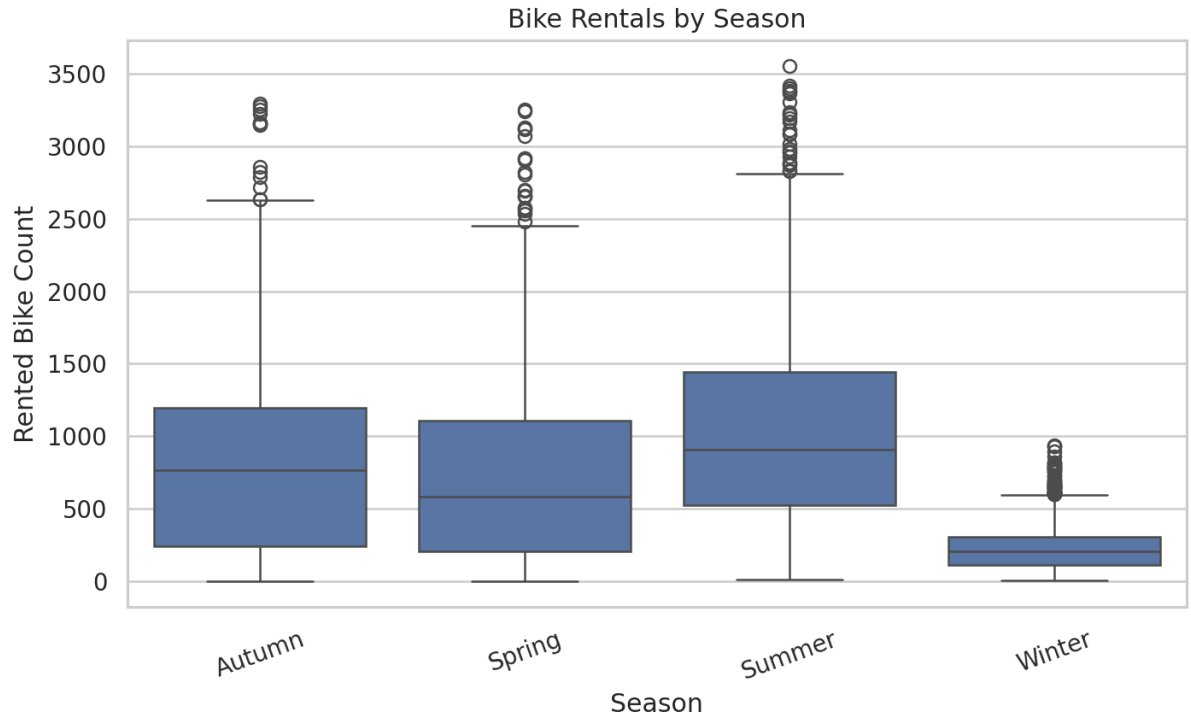
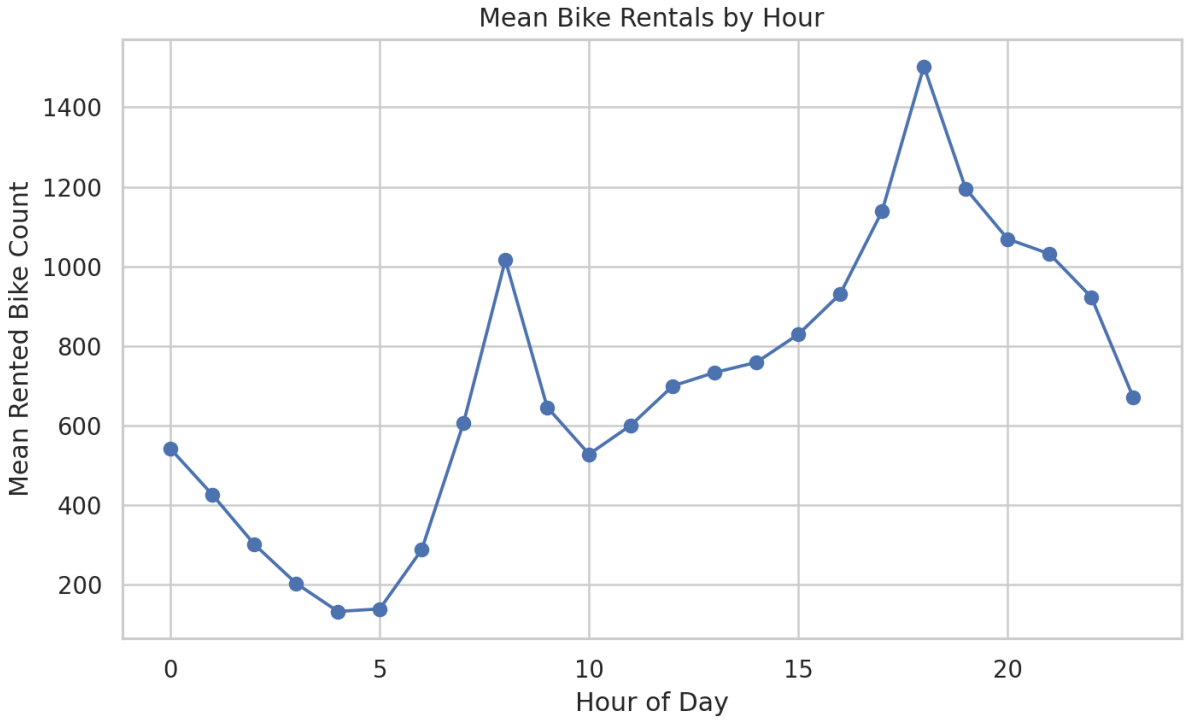


Figure 5. Mean rentals by hour, where hour is available.



C. Simple Linear Regression

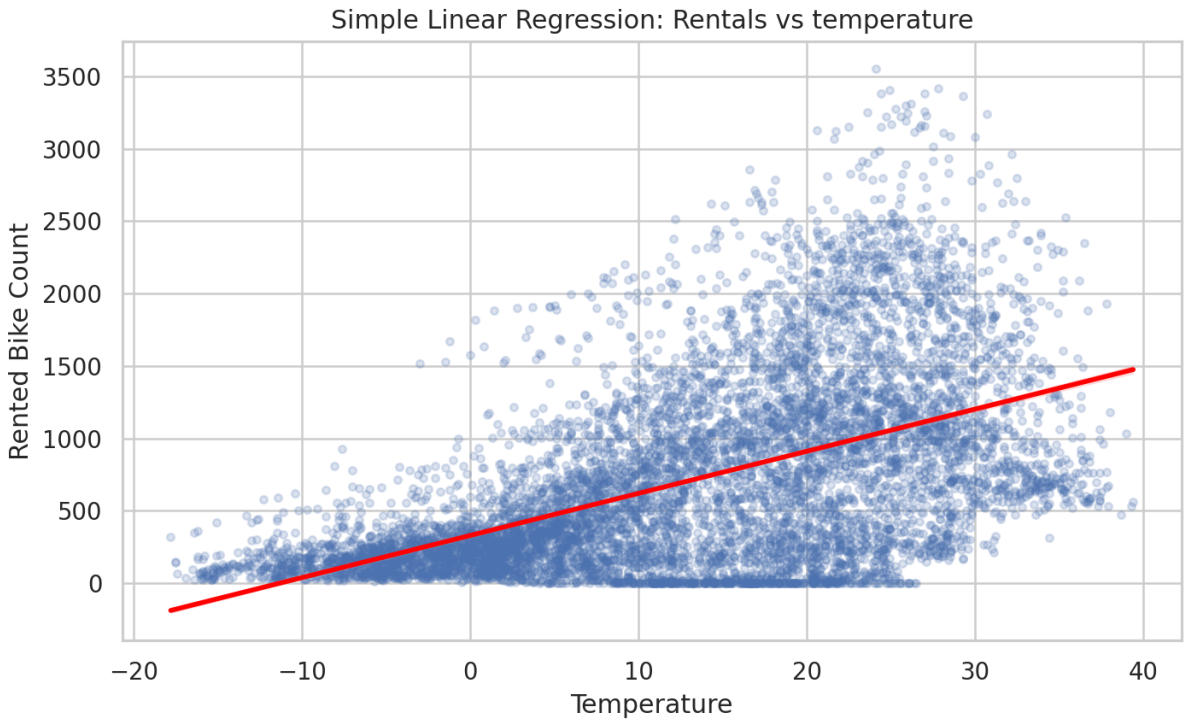
The fitted model is: rental count = intercept + slope × temperature + error. The estimated slope is 29.081, with $p < 0.001$. In the simple model, a one-unit increase in temperature is associated with an estimated change of 29.081 rental units in the mean hourly rental count. The estimated association is positive; its coefficient test gives $p < 0.001$. The model R-squared is 0.290, meaning the model accounts for about 29.0% of the observed variation in rental counts in this sample.

The intercept is the fitted mean rental count when the predictor is zero; this may not be practically meaningful if zero is outside or near the edge of the observed predictor range. R-squared describes the proportion of in-sample response variation accounted for by this one-predictor model, not a causal effect.

Table 4. Simple regression coefficients

Term	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
Intercept	330	8.541	38.63	1.288e-301	313.2	346.7
temperature	29.08	0.4862	59.82	0	28.13	30.03

Figure 6. Simple linear regression fit.



D. Multiple Linear Regression

The multiple regression includes the available quantitative predictors and categorical factors listed in the model formula. Weather, time, and seasonal variables are included because they are plausibly related to rental demand and are explicitly suggested in the assessment. Coefficients are adjusted associations conditional on the other terms in the model.

Model formula: rented_bike_count ~ temperature + humidity + wind_speed + solar_radiation + rainfall + snowfall + hour + visibility + dew_point_temperature + C(seasons) + C(holiday) + C(functioning_day)

The multiple regression uses 8760 complete observations. R-squared = 0.550; adjusted R-squared = 0.550; overall F = 764.812; overall-model p < 0.001. These statistics describe in-sample fit and do not establish causation or guarantee out-of-sample predictive accuracy.

Selected non-intercept terms with p-values below 0.05: C(seasons)[T.Spring] (estimate -135.310, p < 0.001); C(seasons)[T.Summer] (estimate -154.483, p < 0.001); C(seasons)[T.Winter] (estimate -366.054, p < 0.001); C(holiday)[T.No Holiday] (estimate 117.579, p < 0.001); C(functioning_day)[T.Yes] (estimate 932.142, p < 0.001); temperature (estimate 16.071, p < 0.001); humidity (estimate -10.809, p < 0.001); wind_speed (estimate 19.204, p < 0.001)

Statistical significance is not the same as practical importance. A small coefficient can be statistically significant in a large dataset, while a larger estimate can be uncertain. Interpret coefficient size in rental-count units and in the context of the predictor's measurement scale.

Table 5. Multiple-model fit statistics

Model	N	R_squared	Adjusted_R_squared	F_statistic	F_p_value	AIC	BIC
Multiple linear regression	8760	0.5504	0.5497	764.8	0	1.312e+05	1.313e+05

Table 6. Multiple regression coefficients and 95% confidence intervals

Term	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
hour	27.53	0.7347	37.47	4.096e-285	26.09	28.97
C(functioning_day)[T.Yes]	932.1	26.65	34.97	6.033e-251	879.9	984.4
C(seasons)[T.Winter]	-366.1	19.71	-18.57	1.476e-75	-404.7	-327.4
rainfall	-58.48	4.27	-13.69	2.984e-42	-66.85	-50.11
humidity	-10.81	1.03	-10.5	1.256e-25	-12.83	-8.791
solar_radiation	-77.31	7.593	-10.18	3.27e-24	-92.19	-62.43
C(seasons)[T.Spring]	-135.3	13.88	-9.749	2.427e-22	-162.5	-108.1

Term	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
C(seasons)[T.Summer]	-154.5	17.21	-8.976	3.391e-19	-188.2	-120.7
C(holiday)[T.No Holiday]	117.6	21.6	5.442	5.403e-08	75.23	159.9
temperature	16.07	3.662	4.388	1.157e-05	8.892	23.25
wind_speed	19.2	5.095	3.769	0.000165	9.216	29.19
snowfall	32.69	11.21	2.918	0.003534	10.73	54.66
dew_point_temperature	11.16	3.835	2.911	0.003608	3.647	18.68
visibility	0.01029	0.009881	1.042	0.2976	-0.009077	0.02966
Intercept	-82.05	98.61	-0.832	0.4054	-275.3	111.2

E. Statistical Inference

Individual coefficient tests use H0: beta_j = 0 against a two-sided alternative. The coefficient table reports estimates, standard errors, t-statistics, p-values, and 95% confidence intervals. A confidence interval gives a range of parameter values compatible with the fitted model and its assumptions; it does not mean that the fixed parameter has a 95% probability of lying in this particular interval.

The overall F-test evaluates whether the explanatory terms jointly improve on an intercept-only model under the classical linear-model assumptions. The p-value alone does not indicate whether an effect is large enough to matter operationally.

Table 7. Inference for regression coefficients

Term	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
hour	27.53	0.7347	37.47	4.096e-285	26.09	28.97
C(functioning_day)[T.Yes]	932.1	26.65	34.97	6.033e-251	879.9	984.4
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rainfall	-58.48	4.27	-13.69	2.984e-42	-66.85	-50.11
humidity	-10.81	1.03	-10.5	1.256e-25	-12.83	-8.791
solar_radiation	-77.31	7.593	-10.18	3.27e-24	-92.19	-62.43
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F. Matrix Form and Least-Squares Theory

The selected matrix model is y = Xβ + ε.

Here, y is the vector of observed rental counts; X is the design matrix containing a column of ones for the intercept, quantitative predictors, and dummy-variable columns for categorical levels; β is the vector of unknown regression coefficients; and ε is the vector of unobserved errors.

Least-squares estimation minimizes the sum of squared residuals. When X has full column rank, the normal-equation solution is β■ = (X■X)■¹X■y. The code computes this on a reproducible sample and checks it against numpy.linalg.lstsq. If X■X is singular, it uses a pseudoinverse fallback and records that choice.

Table 8. Matrix least-squares coefficient demonstration

Term	Beta_Normal_Equation	Beta_Numpy_Lstsq
Intercept	-87.81	-87.81
C(seasons)[T.Spring]	-133.8	-133.8
C(seasons)[T.Summer]	-37.46	-37.46
C(seasons)[T.Winter]	-375.1	-375.1
C(functioning_day)[T.Yes]	887.5	887.5

Term	Beta_Normal_Equation	Beta_Numpy_Lstsq
temperature	21.61	21.61
humidity	-7.897	-7.897
hour	22.42	22.42

The matrix solution and statsmodels regression use the same least-squares principle. Differences can arise when the sample, formula, reference categories, or design-matrix columns differ.

G. Dummy Variables and ANOVA

Categorical predictors are represented through indicator (dummy) columns. One level is used as the reference category; each reported factor coefficient represents the fitted difference from that reference level, holding other model terms fixed. The default treatment coding in statsmodels uses a reference level determined by the factor's category ordering unless explicitly set.

After controlling for the selected predictors, adding the season factor gives $F = 169.094$, $p < 0.001$. There is evidence that season contributes to the model.

The one-way ANOVA tests whether all seasonal mean rental counts are equal. In a linear-model framework, ANOVA and dummy-variable regression are equivalent ways to represent group mean differences; the overall factor F-test tests the joint contribution of the season indicators.

Table 9. Rental count summaries by season

seasons	count	mean	std
Autumn	2184	819.6	651.1
Spring	2208	730	621.5
Summer	2208	1034	690.2
Winter	2160	225.5	150.4

Table 10. One-way ANOVA for season

index	sum_sq	df	F	PR(>F)
C(seasons)	7.657e+08	3	776.5	0
Residual	2.878e+09	8756		

H. Two-Way ANOVA and Interaction

The interaction model includes season, functioning-day status, and their product term. The interaction F-test compares this model with an additive model. A significant interaction indicates that the seasonal difference is not constant across functioning-day groups on the model's rental-count scale. A nonsignificant test is not proof that all group effects are identical.

The season-by-functioning-day interaction test gives $F = 4.014$, $p = 0.045$. This supports different seasonal patterns by functioning-day status in this model.

Figure 7. Mean rental counts by season and functioning-day status.

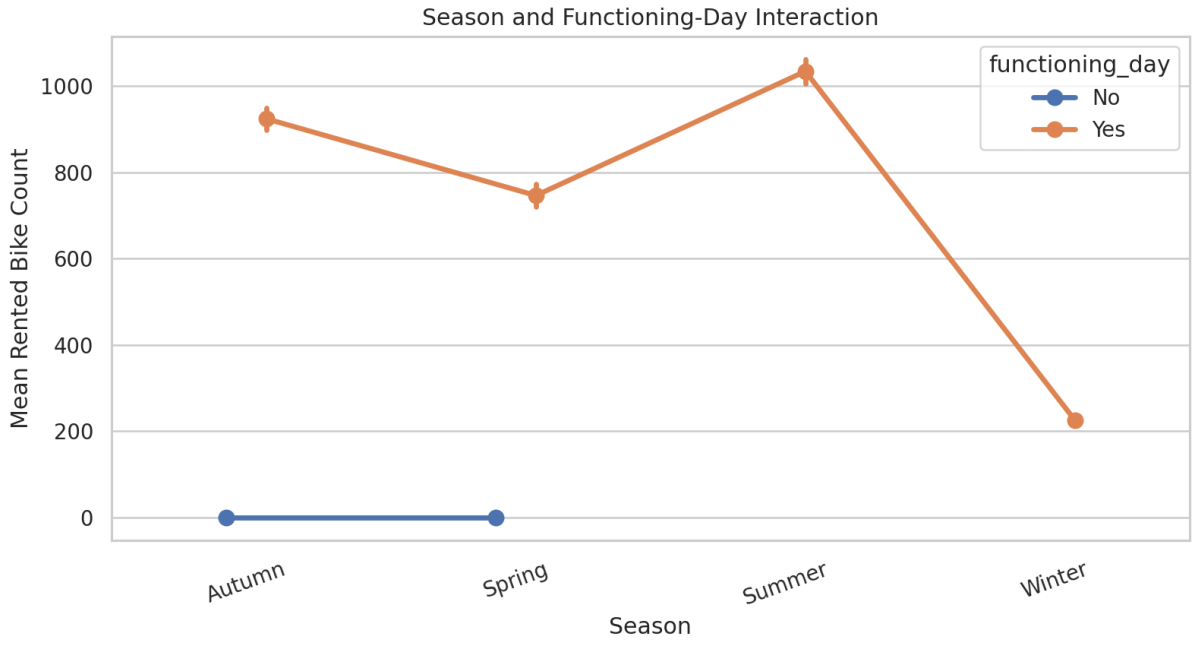


Table 11. Interaction partial F-test

F_statistic	p_value	df_difference
4.014	0.04514	1

I. Model Diagnostics and Validation

The residuals-versus-fitted plot is used to inspect nonlinearity and changing variance; the Q-Q plot assesses residual normality; the scale-location plot examines residual spread; and Cook's distance helps flag potentially influential observations. These plots should be interpreted together rather than as automatic pass/fail rules.

The Breusch-Pagan test suggests possible nonconstant residual variance.

The Shapiro-Wilk test suggests residuals depart from normality.

The RESET test suggests possible model specification issues.

The Durbin-Watson statistic is 0.509. Values near 2 are consistent with little first-order residual autocorrelation, but the statistic is not a complete test of independence. Because these data are hourly and time-ordered, residual dependence is a meaningful concern. Ordinary regression p-values may be unreliable if temporal dependence is substantial.

379 observations were flagged by the exploratory rule of Cook's distance greater than 4/n or absolute internally studentized residual greater than 3. Flagged cases should be investigated rather than automatically deleted.

Table 12. Diagnostic tests

Test	Statistic	p_value	Interpretation_rule
Breusch-Pagan test for heteroscedasticity	844.3	3.651e-171	Small p-value suggests nonconstant variance
Shapiro-Wilk residual normality test	0.947	3.935e-39	Small p-value suggests non-normal residuals
Ramsey RESET specification test	1382	5.171e-281	Small p-value may indicate misspecification
Durbin-Watson statistic	0.5095		Near 2 suggests limited first-order residual autocorrelatio

Figure 8. Residuals versus fitted values.

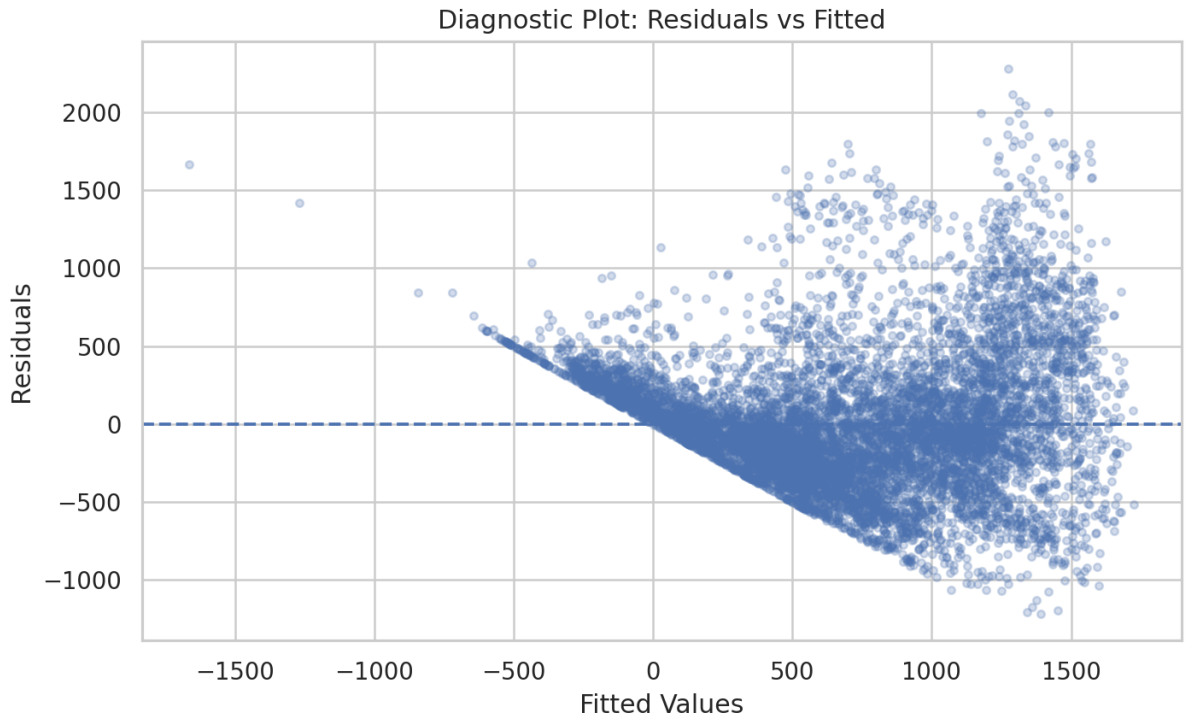


Figure 9. Normal Q-Q plot.

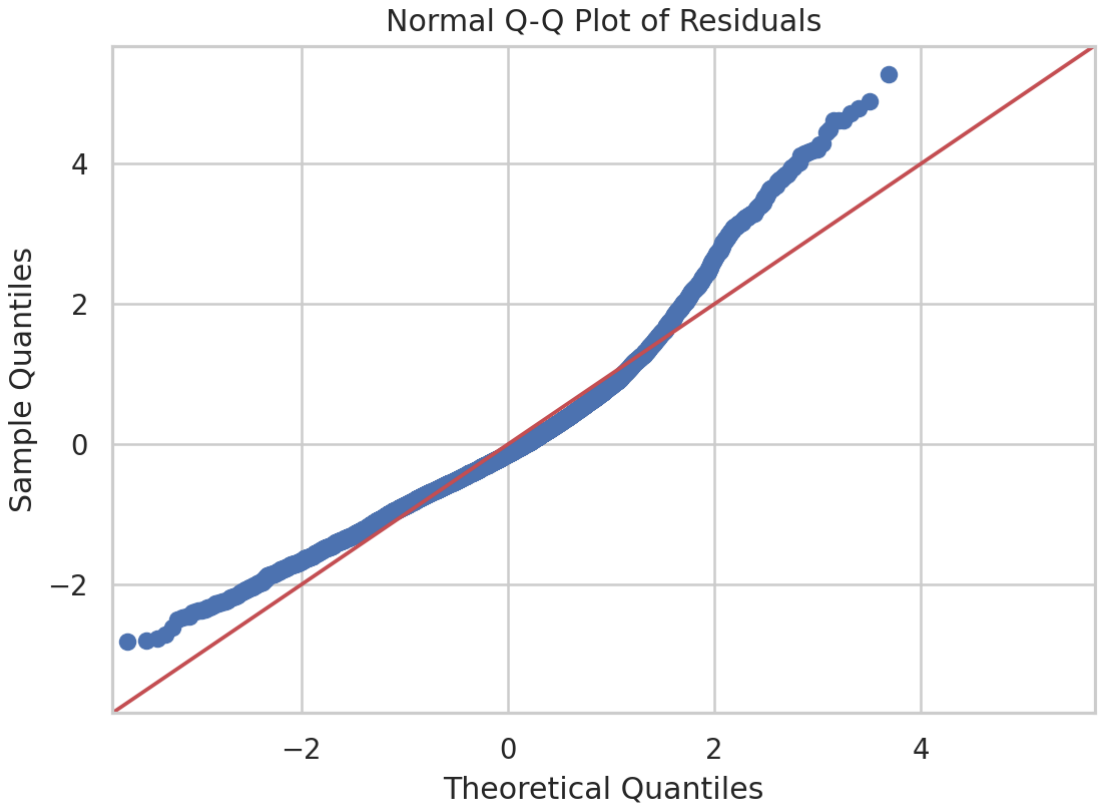


Figure 10. Scale-location plot.

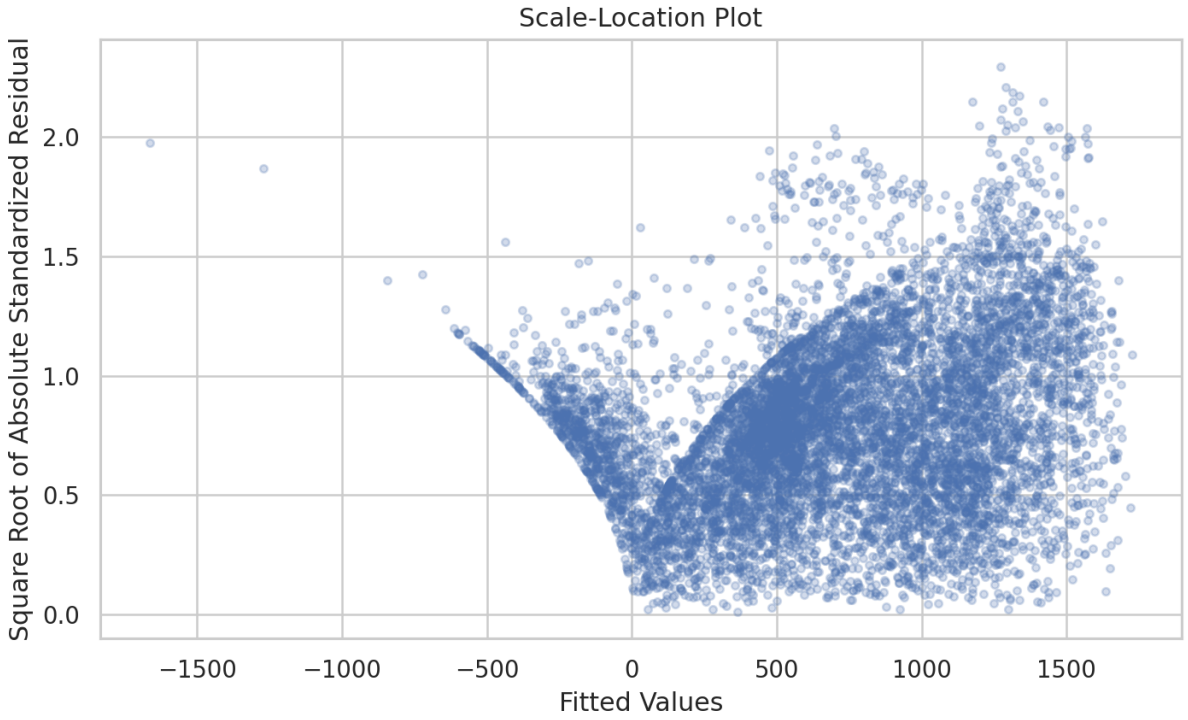
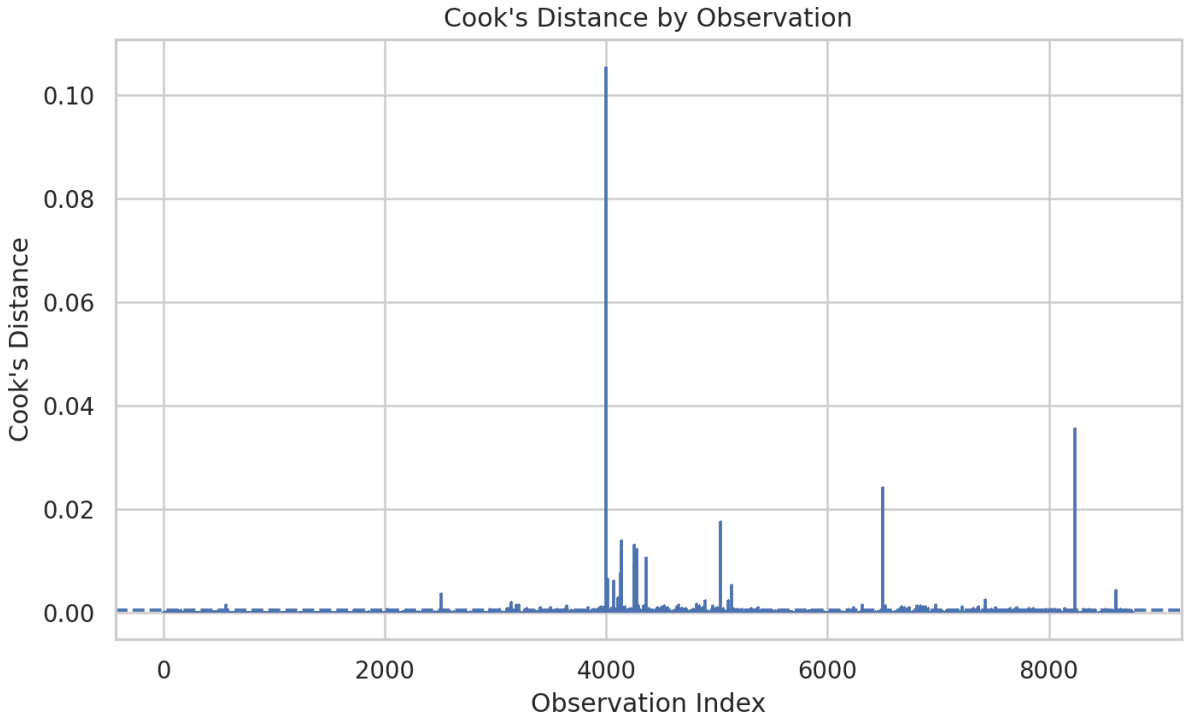


Figure 11. Cook's distance.



Transformation and alternative specification

Rental counts can be skewed and may show nonconstant variance. The log1p response model is provided as a sensitivity analysis, not as an automatic final model. Its R-squared and AIC are on a different response scale and should not be directly compared with the original-response model as if they were identical criteria. If prediction is the main goal, a Poisson or negative-binomial model and time-aware validation may be useful extensions, although they are outside the main linear-model assessment.

J. Final Model and Statistical Communication

The selected primary model is the multiple linear regression using the available weather, time, and categorical predictors. It is chosen because it directly addresses the research question, adjusts for multiple factors, and supports coefficient inference, confidence intervals, dummy-variable interpretation, and model diagnostics. Its limitations are the linear functional form, possible count-data skewness, potential temporal dependence, and possible heteroscedasticity. The diagnostic findings should be considered before using its p-values for strong decisions.

In nontechnical terms, the analysis estimates how bike rentals tend to change alongside weather and operating conditions. The regression results identify which associations are most evident in this dataset, while the confidence intervals and diagnostics indicate how much uncertainty remains. These observational data can inform planning, but they do not prove that changing a weather or operating condition would cause a particular change in rentals.

Conclusion: The research question is addressed by estimating the simple and multiple regression relationships and testing seasonal and functioning-day differences. The actual direction, size, and strength of each association should be reported using the generated tables, confidence intervals, and diagnostic results rather than assuming any particular result in advance.

References

UCI Machine Learning Repository. Seoul Bike Sharing Demand Dataset.
https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand

Seoul Bike Sharing Demand dataset distributed as SeoulBikeData.csv. Python libraries used: pandas, NumPy, SciPy, statsmodels, patsy, Matplotlib, seaborn, and ReportLab.